Neural Information Retrieval at Scale

Section 1: Comparative Ingestion Architectures

1.1 Ingestion Vulnerabilities

Modern retrieval pipelines depend critically on high-fidelity document ingestion. When parsing scientific papers, reports, or legal filings, multi-column layouts present a significant challenge for naive parsers. If an engine reads text streams linearly by vertical position without clustering bounding boxes into spatial columns, sentences from opposing columns get horizontally interleaved. This corrupts semantic continuity before chunking and embedding generation even begin.

1.2 Spatial Geometry Solutions

To eliminate the reading order trap, modern ingestion engines cluster text spans into discrete spatial bounding boxes. By evaluating horizontal separation bands and grouping text into column clusters, the parser preserves the author's intended reading order. Spatial coordinate reconstruction guarantees that Column 1 is read fully before Column 2, protecting the semantic integrity of downstream chunks.

Section 2: Empirical Parser Comparison

The table below summarizes the key architectural distinctions between PyMuPDF and pypdf. In production RAG systems, licensing, throughput, and layout retention determine whether a parser fits enterprise requirements.

Feature / Metric	PyMuPDF (fitz)	pypdf
Underlying Core	Native C (MuPDF)	Pure Python
Processing Speed	Ultra-High (2-15 ms)	Moderate (30-150 ms)
Bounding Box Metadata	Native coordinates	Basic text streams
Enterprise License	AGPL-3.0 / Commercial	Apache 2.0 Permissive